Amazon India Market Entry Analysis

Leveraging Amazon Brazil's Data to Drive Strategic Decisions for the Indian Market

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Prepared By : Yamini Thakur (Next Leap : Cohort 10)
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Executive Summary

Amazon has established a strong foothold in global e-commerce markets including the United States, Europe, and Asia. In Brazil, Amazon has successfully connected small and medium businesses with millions of customers, creating a robust and growing marketplace.

Given the structural similarities between Brazil and India including large and diverse populations, a growing middle class, price-sensitive consumers, and expanding digital payment ecosystems, Brazil's e-commerce data presents a valuable opportunity to inform Amazon India's market strategy.

This report presents a comprehensive analysis of Amazon Brazil's transactional data spanning customer behavior, payment preferences, product categorization, seasonal trends, and revenue patterns. The analysis is structured across three sections:

- **Analysis I — Payment standardization, product pricing, and data quality**
- **Analysis II — Customer segmentation, category revenue, and price distribution**
- **Analysis III — Seasonal trends, growth patterns, and payment method performance**

Each section contains SQL-backed findings, business implications, recommended actions, and India-specific caveats to ensure that insights from Brazil are thoughtfully adapted for the Indian market.

Objective

To identify trends, customer behaviors, and preferences from Amazon Brazil's data that can be leveraged to help Amazon India make informed decisions, enhance customer experience, and seize new market opportunities.

Analysis I

Q1: Amazon India wants to standardise payments to simplify financial records. Average payment values are taken to solve the problem of payment standardization.

Query | Query History

```
1
2   -- Analysis - I
3   -- To simplify its financial reports, Amazon India needs to standardize payment values.
4   -- Round the average payment values to integer (no decimal) for each payment type and display the results sorted in ascending order.
5
6   -- Output: payment_type, rounded_avg_payment
7   select
8       payment_type,
9       round(avg(payment_value)) as rounded_avg_paymnet
10  from payments
11      group by 1
12      order by 2
13
14
15
16
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1

	payment_type character varying	rounded_avg_paymnet numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Q1	Recommendations
FINDING	The data shows that voucher is the least used payment type and credit card is mostly used for payments.
IMPLICATION	• people are using voucher for small payments only • they do not get much vouchers for making payments • number of payments made by credit cards are comparatively lower resulting in higher average. • they prefer to use EMI options or credit options to buy products
ACTION	Amazon can improve EMI options and streamline payment by credit card options.
CAVEAT	For India, credit card payments might not be as prevalent as in Brazil We will have to look into the most prevalent methods of payments for India.

Q2: Amazon India want to understand which payment type is most prevalent.

Query | Query History

```
-- To refine its payment strategy, Amazon India wants to know the distribution of orders by payment type.
-- Calculate the percentage of total orders for each payment type, rounded to one decimal place,
-- and display them in descending order

-- Output: payment_type, percentage_orders
select
    payment_type,
    round(count(order_id)*100.0/(select count(order_id) from payments),1) as percentage_orders
from payments
    group by payment_type
    order by 2 desc
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	payment_type character varying	percentage_orders numeric
1	credit_card	73.9
2	boleto	19.0
3	voucher	5.6
4	debit_card	1.5
5	not_defined	0.0

Q2	Recommendations
FINDING	The data shows that debit card is the least used payment type and credit card is mostly used for payments, similar to what we got in the last question.
IMPLICATION	• It is easier and hassle free to make payments via credit card. • people do not have immediate money in banks rather they have money in hand so they prefer to pay in cash as compared to debit card. • people face resistance in making payments by debit card.
ACTION	• Amazon India should look at the friction points which the people are experiencing while making payments through dr cards. • Is it that they have low money in bank or is the process of making payment through dr card not quick, are there bank server issues • It should also look at adding more EMI options and see what feature of credit card payment type is making people choose that option. Give emphasis on those features and try to incorporate them in other payment options if possible.
CAVEAT	As compared to Brazil 's financial ecosystem, India has robust UPI system, people might prefer to make payments via UPI as it is easier and quick. Amazon India should incorporate UPI option and see if payment by credit card is still the most prevalent option.

Q3: Amazon India seeks to create targeted promotions for products within specific price ranges.

Query | Query History

```
25
26
27  -- Amazon India seeks to create targeted promotions for products within specific price ranges.
28  -- Identify all products priced between 100 and 500 BRL that contain the word 'Smart' in their name.
29  -- Display these products, sorted by price in descending order.
30
31  -- Output: product_id, price
32  select
33      p.product_id,
34      oi.price
35  from order_items oi
36      join products p
37      on oi.product_id = p.product_id
38  where (oi.price>100 and oi.price <500) and p.product_category_name like 'smart%'
39  order by 2 desc
40
41
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 33

	product_id character varying	price numeric (10,2)
1	1df1a2df8ad2b9d3aa49fd851e3145ad	439.99
2	7debe59b10825e89c1cbcc8b190c85...	349.99
3	ca86b9fe16e12de698c955aedff0aea2	349.00
4	ca86b9fe16e12de698c955aedff0aea2	349.00
5	0e52955ca8143bd179b311cc454a6c...	335.00
6	7aeaa8f3e592e380c420e8910a7172...	329.90
7	7aeaa8f3e592e380c420e8910a7172...	329.90
8	7aeaa8f3e592e380c420e8910a7172...	329.90
9	7aeaa8f3e592e380c420e8910a7172...	329.90
10	7aeaa8f3e592e380c420e8910a7172...	329.90

Total rows: 33 | Query complete 00:00:00.125

Q3	Recommendations
FINDING	There are 33 product ids which have smart in their name and price varies from 102 - 440
IMPLICATION	As India is a price-sensitive market so one of the categories is to choose products in the price range of 100 - 500 and products which specifically mention smart in their name is one category to target as India is also tech savvy market.
ACTION	Amazon India can focus on promoting these 33 categories It can also look for other combinations of product name and price range like ' Combo under 100', 'Top quality under 500'
CAVEAT	Amazon India need to check the order values and frequency of purchase for these products, out of 33 product ids, it should focus more on such product ids which are more frequently purchased and have high order values.

Q4: Amazon India wants to identify seasonal sales patterns; to understand this, it focuses on the most successful months.

Query | Query History

```
95
96  -- To identify seasonal sales patterns, Amazon India needs to focus on the most successful months.
97  -- Determine the top 3 months with the highest total sales value, rounded to the nearest integer.
98
99  -- Output: month, total_sales
100 select
101     extract(month from o.order_purchase_timestamp) as month_of_year,
102     round(sum(p.payment_value),2) as total_sales
103 from orders o
104     join payments p
105     on o.order_id = p.order_id
106 group by 1
107 order by 2 desc
108 limit 3
109
110
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 3

	month_of_year numeric	total_sales numeric
1	5	1746900.97
2	8	1696821.64
3	7	1658923.67

Q4	Recommendations
FINDING	• The top 3 months which have highest total sales are May, July and August, In India, it corresponds to later summer and early rainy season. • Whereas in Brazil these months correspond to winter season (June to September). So based on the results, it can be seen that people tend to make more purchases during the winter seasons in Brazil.
IMPLICATION	• One possible driver for the high sales amount for these months can be amazon specific promotions like Early winter sales offers in May or Late winter sales offers in August. • Mid-year winter break runs from early July to late July in Brazil, resulting in increased holiday and travel expenses.
ACTION	• Amazon India should also look at the products bought during these months and see if there are any products which are more frequently bought during these months only and if the sale of those products declined with change in the season. • Does the high sales figure attribute to any promotion campaigns run by amazon during these months?
CAVEAT	• In Indian context, Amazon might see the same result as India has summer holidays in July which might see the increased spending during holiday seasons. • Still, India has specific festivals and culture has significant effect on consumer spending like strong sales figures in the months of October and November due to Diwali and Dusshera when Indian spend the most on everything. • Amazon India should gauge the cultural angle to derive insights on season wise sales in India.

Q5: Amazon India is interested in product categories with significant price variations

Query | Query History

```
105
106
107  -- Amazon India is interested in product categories with significant price variations.
108  -- Find categories where the difference between the maximum and minimum product prices is greater than 500 BRL.
109
110  -- Output: product_category_name, price_difference
111  select
112      p.product_category_name,
113      (max(oi.price)-min(oi.price)) as price_difference
114  from products p
115      join order_items oi
116      on p.product_id = oi.product_id
117  group by 1
118  having (max(oi.price)-min(oi.price)) >500
119  order by 2 desc
120
121
122
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 57

	product_category_name character varying	price_difference numeric
1	utilidades_domesticas	6731.94
2	pcs	6694.50
3	artes	6495.50
4	eletroportateis	4792.50
5	instrumentos_musicais	4394.97
6	consoles_games	4094.81
7	esporte_lazer	4054.50
8	relogios_presentes	3990.91
9	[null]	3977.00

Total rows: 57 | Query complete 00:00:00.382

Q5	Recommendations
FINDING	• There are 57 product categories which have price difference greater than 500 BRL
IMPLICATION	• These products categories serve different types of customers, budget buyers and premium buyers. • High price variation also indicates that the product category has local sellers and brand sellers • It can also indicate inconsistent cataloguing; total unrelated products must have been grouped in one category resulting in high price variation.
ACTION	• Amazon India can cater to both budget shoppers and premium buyers of customers with a product category by showing the price range based on past customer spending • They can also identify if there are any cheap variations of the product which are hurting the sales of premium product. • They can decide on which sellers to onboard or prioritise.
CAVEAT	• Should all 57 categories be treated the same way or should Amazon India prioritise categories with the highest price difference first? • In India, which of these 57 categories are even relevant? A category popular in Brazil may have no demand in India at all.

Q6: To enhance the customer experience, Amazon India wants to find which payment types have the most consistent transaction amounts.

Query | Query History

```
69
70
71
72  -- To enhance the customer experience, Amazon India wants to find which payment types have the most consistent transaction amounts.
73  -- Identify the payment types with the least variance in transaction amounts, sorting by the smallest standard deviation first.
74
75  -- Output: payment_type, std_deviation
76  select
77      payment_type,
78      round(stddev(payment_value),2) as std_deviation
79  from payments
80  group by 1
81  order by 2 asc
82
83
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 5 | Page No: 1 of 1

	payment_type character varying	std_deviation numeric
1	not_defined	0.00
2	voucher	115.52
3	boleto	213.58
4	credit_card	222.12
5	debit_card	245.79

Q6	Recommendations
FINDING	• Voucher is the payment type with least variation, which means the payments made by voucher are almost of same value. • Payment by Debit card has highest standard deviation, indicating different number of purchases made.
IMPLICATION	• People are not intentionally making a purchase rather they are buying because they have received a voucher. Therefore, amounts are closer to each other, which means people only spent an amount closer to or equal to the voucher amount. • The vouchers are not growing the cart size; they are only triggering a purchase.
ACTION	• Amazon India can use vouchers as a strategy. • They should design vouchers that require a minimum spend above the voucher value to redeem like get Rs 200 voucher on purchases above Rs 1000, this forces the customer to spend intentionally rather than just redeeming the voucher amount.
CAVEAT	In India, voucher behaviour may differ significantly from Brazil. Indian customers are highly coupon-driven but tend to combine vouchers with additional spending due to the strong cashback culture on platforms like Paytm and PhonePe. Amazon India should test whether vouchers cap spending the same way as in Brazil or whether Indian customers spend beyond the voucher value, before deciding on voucher denomination strategy.

Q7: Amazon India wants to identify products that may have incomplete name in order to fix it from their end.

Query | Query History

```
82
83  -- Amazon India wants to identify products that may have incomplete name in order to fix it from their end.
84  -- Retrieve the list of products where the product category name is missing or contains only a single character.
85
86  -- Output: product_id, product_category_name
87  select
88      product_id,
89      product_category_name
90  from products
91  where product_category_name is null or length(product_category_name) = 1;
92
93
94
95
96
```

Data Output | Messages | Notifications

Showing rows: 1 to 614

	product_id [PK] character varying	product_category_name character varying
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdcbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]
8	794de06c32a626a5692ff50e4985d3...	[null]
9	7af3e2da474486a3519b0cba9dea8...	[null]
10	629beb8e7317703dcc5f35b5463fd2...	[null]
11	3a78f64aac654298e4b9aff32fc21818	[null]

Total rows: 614 | Query complete 00:00:00.121

Q7	Recommendations
FINDING	There are 614 products which are not categorised properly.
IMPLICATION	It indicates sellers are not listing their products properly, which might result in low sales of the products, eventually losing sales and trust on Amazon.
ACTION	• While searching on Amazon, most people do not search specific names of the products like “Nike Men Air Zoom Pegasus 41 Running Shoes”, they will search for “Running Shoes” or “Nike shoes”, this product comes in “Sportswear and Footwear category”. But, if this product is not listed under this category, then unless someone types in the full name of this shoe, no one is ever going to find it on Amazon and there will be no sale. • Amazon must create a mandatory step in the process of adding a new product on the page, it should not accept the product unless categorised. • Amazon should also create standardised categories across the app.
CAVEAT	In India, most sellers in Amazon are small sellers who do not understand much about the categorisation of products, so they might end up putting a product in random category just to move past the mandatory step. So, amazon should also recommend possible categories for a product item so that a seller can place his product in a category without getting confused. For example: Seller’s product is “Coconut oil”, the categories can be 1. Food and groceries: If it is edible 2. Health & Wellness: If it is used as health supplement 3. Beauty & Personal Care: If it is used for skin moisturising

Analysis II

#1 Amazon India wants to understand which payment types are most popular across different order value segments (e.g., low, medium, high). Segment order values into three ranges: orders less than 200 BRL, between 200 and 1000 BRL, and over 1000 BRL. Calculate the count of each payment type within these ranges and display the results in descending order of count
Output: order_value_segment, payment_type, count

Query | Query History

```
91
92  -- Analysis - II
93
94  -- #1 Amazon India wants to understand which payment types are most popular across different order value segments
95  -- (e.g., low, medium, high).
96  -- Segment order values into three ranges: orders less than 200 BRL, between 200 and 1000 BRL, and over 1000 BRL.
97  -- Calculate the count of each payment type within these ranges and display the results in descending order of count
98
99  -- Output: order_value_segment, payment_type, count
100
101     select
102         case when payment_value <200
103              then '1 Less than 200 BRL'
104              when payment_value <1000
105              then '2 Between 200 and 1000 BRL'
106              when payment_value >1000
107              then '3 Over 1000 BRL'
108         end as order_value_segment,
109         payment_type,
110         count(payment_value)
111     from payments p
112     group by 1,2
113     order by 2,1 asc
```

Data Output | Messages | Notifications

Showing rows: 1 to 13 | Page No: 1

	order_value_segment text	payment_type character varying	count bigint
9	3 Over 1000 BRL	debit_card	15
10	1 Less than 200 BRL	not_defined	3
11	1 Less than 200 BRL	voucher	5476
12	2 Between 200 and 1000 B...	voucher	286

Total rows: 13 | Query complete 00:00:00.116

Data Output | Messages | Notifications

	order_value_segment text	payment_type character varying	count bigint
1	1 Less than 200 BRL	boleto	16444
2	2 Between 200 and 1000 B...	boleto	3162
3	3 Over 1000 BRL	boleto	178
4	1 Less than 200 BRL	credit_card	60548
5	2 Between 200 and 1000 B...	credit_card	15303
6	3 Over 1000 BRL	credit_card	944
7	1 Less than 200 BRL	debit_card	1287
8	2 Between 200 and 1000 B...	debit_card	227
9	3 Over 1000 BRL	debit_card	15
10	1 Less than 200 BRL	not_defined	3
11	1 Less than 200 BRL	voucher	5476
12	2 Between 200 and 1000 B...	voucher	286
13	3 Over 1000 BRL	voucher	13

Analysis II Q1	Recommendations
FINDING	• There are 4 payment_types 'boleto', 'credit_card', 'debit_card', 'voucher' and payment_type for 3 payments is not defined. • For all order_value_segments, use of credit_card is more prevalent than any other payment_type. • Boleto is the 2nd most preferred payment_type and debit_card is the least preferred one.
IMPLICATION	There is no effect of different order_value_segments on the payment_type which people use. It means if a customer is using credit card for purchases worth 200BRL or less, then they are likely to use the same for purchases worth 1000 BRL or more. So, we cannot say that one payment_type is popular in one order_value_segment.
ACTION	• Amazon can create a different order_value_segment and see if changing the segments brings out a significant result. • It can focus on the popularity of credit_card and look into the reasons and implement the same for other payment_types to increase their popularity.
CAVEAT	• Boleto is an official, cash-based voucher payment system regulated by the Central Bank of Brazil. It generates a barcode and a reference number that consumers use to pay for goods, services, and bills via banking apps, ATMs, or in person at thousands of lottery agents and convenience stores. In India, **Bharat Connect** (formerly known as the **Bharat Bill Payment System** or **BBPS**) is one such payment_method which is seldom used in India for consumer shopping in Amazon. • Amazon India will have to look at Indian specific payment_methods and create segments based on India's propensity to consume and the type of products India is purchasing. For example: The least amount and the most amount spent on an e-commerce purchase also changes based on the items purchased, if it is groceries, it can range from 100 – 500, in case of electronics, it will start with 1000 itself.

#2 Amazon India wants to analyse the price range and average price for each product category.
Calculate the minimum, maximum, and average price for each category, and list them in descending order by the average price.
Output: product_category_name, min_price, max_price, avg_price

Query | Query History

```
115
116  -- #2 Amazon India wants to analyse the price range and average price for each product category.
117  -- Calculate the minimum, maximum, and average price for each category, and list them in descending order by the average price.
118
119  -- Output: product_category_name, min_price, max_price, avg_price
120  select
121      p.product_category_name,
122      min(oi.price) as minimum_price,
123      max(oi.price) as maximum_price,
124      round(avg(oi.price),2) as average_price
125  from order_items oi
126  join products p
127  on oi.product_id = p.product_id
128  group by p.product_category_name
129  order by 4 desc
130
```

Data Output | Messages | Notifications

Showing rows: 1 to 79 | Page No: 1 of 1

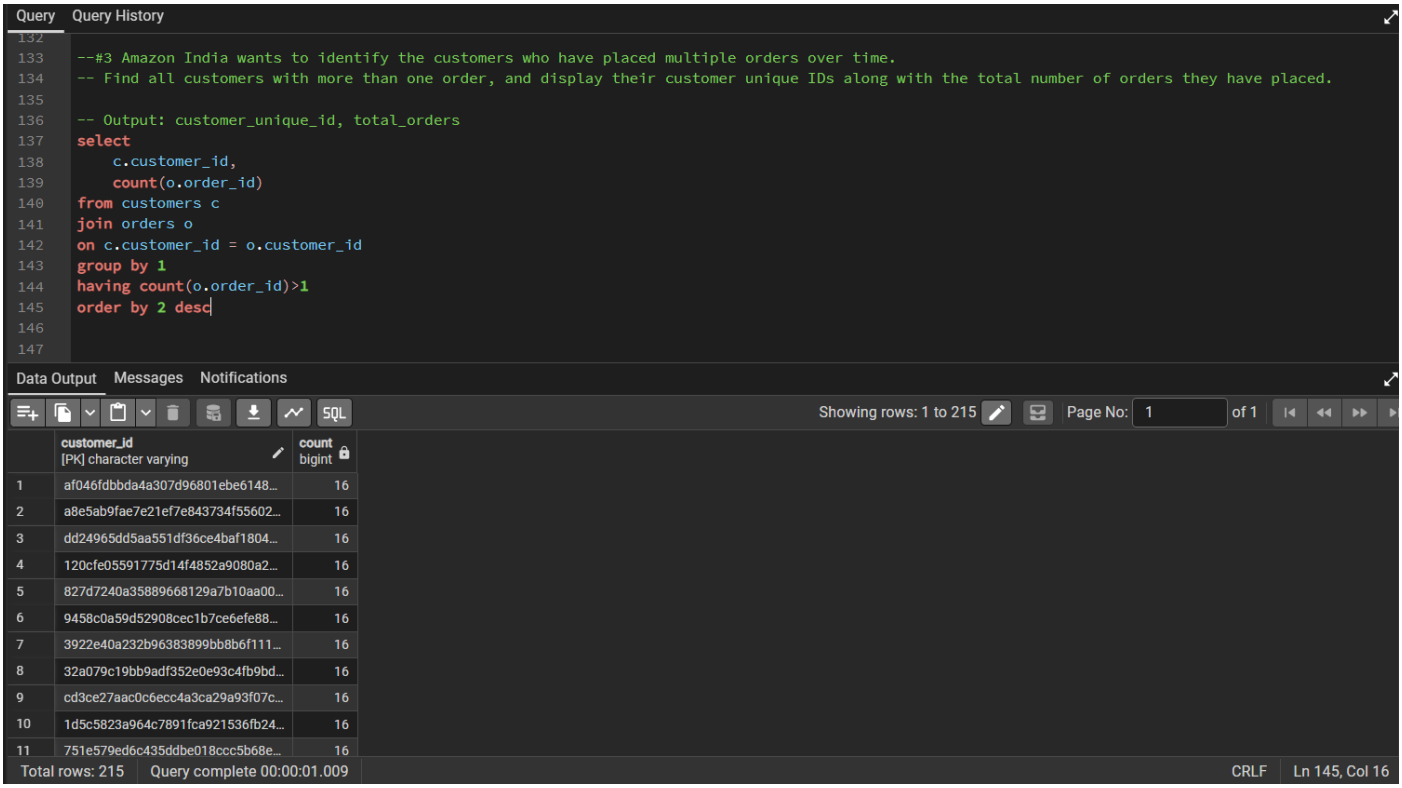
	product_category_name character varying	minimum_price numeric	maximum_price numeric	average_price numeric
1	pcs	34.50	6729.00	1098.34
2	portateis_casa_forno_e_cafe	10.19	2899.00	624.29
3	eletrodomesticos_2	13.90	2350.00	476.12
4	agro_industria_e_comercio	12.99	2990.00	341.66
5	instrumentos_musicais	4.90	4399.87	281.62
6	eletroportateis	6.50	4799.00	280.78
7	portateis_cozinha_e_preparadores_de_aliment...	17.42	1099.00	264.57
8	telefonia_fixa	6.00	1790.00	225.69
9	construcao_ferramentas_seguranca	8.90	3099.90	208.99
10	relogios_presentes	8.99	3999.90	200.91
11	climatizacao	10.90	1599.00	185.27

Total rows: 79 | Query complete 00:00:00.283 | CRLF

Analysis II Q2	Recommendations
FINDING	• Top 5 product_category with top 5 average price are "pcs", "portateis_casa_forno_e_cafe", "eletrodomesticos_2", "agro_industria_e_comercio", "instrumentos_musicais", which might mean: PCs, home appliances (ovens/coffee machines), white goods/appliances, agro-industrial/commercial equipment, musical instruments. • Bottom 5 product_categories with bottom 5 average price are "cds_dvds_musicais", "fraldas_higiene", "flores", "casa_conforto_2", "f", which probably mean: CDs/DVDs, diapers/hygiene, flowers, home comfort items, and one undefined/corrupted category.
IMPLICATION	Eectronics and machinery products with high average price are rarely and have high margins per unit bought whereas consumer goods with low average price are the most frequent purchases with low margins per unit.
ACTION	Amazon can focus on promoting the sales of high margin product categories. • Detailed specs, comparison charts, expert reviews • Financing options like EMI, no-cost EMI, credit card offers (since ticket size is high) • After-sales Service like Warranty, installation support, return policy clarity • Retarget customers who searched/browsed but didn't buy (long decision cycles) Amazon can opt this strategy for low margin, high frequency product categories. • Make money through repeat purchases, not one big sale • Encourage auto-replenishment (e.g., diapers every month)

	• Fast delivery matters more. • Combine with other small items to increase basket size • Show on homepage, "frequently bought together," cart add-ons.
CAVEAT	Amazon will have to check whether the same product categories behave in the same manner in India as they do in Brazil or are there any product categories which are more prominent than what we saw in this analysis.

#3 Amazon India wants to identify the customers who have placed multiple orders over time.
Find all customers with more than one order, and display their customer unique IDs along with the total number of orders they have placed.
Output: customer_unique_id, total_orders



```
--#3 Amazon India wants to identify the customers who have placed multiple orders over time.
-- Find all customers with more than one order, and display their customer unique IDs along with the total number of orders they have placed.

-- Output: customer_unique_id, total_orders
select
    c.customer_id,
    count(o.order_id)
from customers c
join orders o
on c.customer_id = o.customer_id
group by 1
having count(o.order_id)>1
order by 2 desc
```

Data Output | Messages | Notifications — Showing rows: 1 to 215 — Page No: 1 of 1

	customer_id [PK] character varying	count bigint
1	af046fdbbda4a307d96801ebe6148...	16
2	a8e5ab9fae7e21ef7e843734f55602...	16
3	dd24965dd5aa551df36ce4baf1804...	16
4	120cfe05591775d14f4852a9080a2...	16
5	827d7240a35889668129a7b10aa00...	16
6	9458c0a59d52908cec1b7ce6efe88...	16
7	3922e40a232b96383899bb8b6f111...	16
8	32a079c19bb9adf352e0e93c4fb9bd...	16
9	cd3ce27aac0c6ecc4a3ca29a93f07c...	16
10	1d5c5823a964c7891fca921536fb24...	16
11	751e579ed6c435ddbe018ccc5b68e...	16

Total rows: 215 | Query complete 00:00:01.009 | CRLF | Ln 145, Col 16

Analysis II Q3	Recommendations
FINDING	The data represents 98348 customers, out of which only 215 customers have made a purchase more than once, with 16 being the highest frequency.
IMPLICATION	There are a lot of one-time purchasers.
ACTION	Amazon should determine the reasons as to why the customer did not return after making the first purchase. • Post-purchase engagement: Follow-up email/notification a few days after delivery asking for review, suggesting related products. • Second-purchase incentive: "Get 10% off your next order"; time-bound offers create urgency • Personalized recommendations: Based on their one purchase, show complementary products • Remove friction: Make sure delivery and return experience was smooth, since a bad first experience guarantees no second purchase For all the customers who have made frequent purchases • Loyalty program: Points, cashback, or tier-based perks, reward every additional purchase. • Subscribe & Save: For repeat-purchase categories, lock them into a recurring habit • Let loyal customers see deals/sales before everyone else • Exclusive support: Faster customer service, free returns, priority delivery • Personalization: Use their purchase history to recommend before they even search, reduce effort to buy again

#4 Amazon India wants to categorize customers into different types ('New – order qty. = 1' ; 'Returning' –order qty. 2 to 4; 'Loyal' – order qty. >4) based on their purchase history.

Use a temporary table to define these categories and join it with the customers table to update and display the customer types.

Output: customer_unique_id, customer_type

Query | Query History

```
--#4 Amazon India wants to categorize customers into different types
-- ('New - order qty. = 1' ;  'Returning' -order qty. 2 to 4;  'Loyal' - order qty. >4) based on their purchase history.
-- Use a temporary table to define these categories and join it with the customers table to update and display the customer types.

-- Output: customer_unique_id, customer_type
with temp_table as
    (
        select customer_id,
               count(order_id) as no_of_purchases,
               case when count(order_id) = 1
               then 'New'
               when count(order_id)<4
               then 'Returning'
               when count(order_id)>4
               then 'Loyal'
               end as type_of_customer
        from orders
        group by customer_id
    )
select c.customer_unique_id, t.type_of_customer
from customers c
join temp_table t
on c.customer_id = t.customer_id
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 1000 | Page No: 1 of 99

	customer_unique_id character varying	type_of_customer text
1	248ffe10d632bebe4f7267f1f44844c9	New
2	b0015e09bb4b6e47c52844fab5fb66...	New
3	4893ad4ea28b2c5b3ddf4e82e79db9...	New
4	0b83f73b19c2019e182fd552c048a2...	New

Total rows: 98348 | Query complete 00:00:01.234 | CRLF

Analysis II Q4	Recommendations
FINDING	The data represents 98348 customers, majority of which are new.
IMPLICATION	There are a lot of one-time purchasers.
ACTION	Same as in **Analysis II Q3**

#5 Amazon India wants to know which product categories generate the most revenue.
Use joins between the tables to calculate the total revenue for each product category.
Display the top 5 categories.
Output: product_category_name, total_revenue

Query | Query History

```
169
170  -- #5 Amazon India wants to know which product categories generate the most revenue.
171  -- Use joins between the tables to calculate the total revenue for each product category.
172  -- Display the top 5 categories.
173
174  -- Output: product_category_name, total_revenue
175  select
176      product_category_name,
177      sum(payment_value) as total_revenue
178  from products p
179  join order_items oi
180  on p.product_id = oi.product_id
181  join payments ps
182  on oi.order_id = ps.order_id
183  group by 1
184  order by 2 desc
185  limit 5
186
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 5

	product_category_name character varying	total_revenue numeric
1	cama_mesa_banho	1706393.64
2	beleza_saude	1656408.68
3	informatica_acessorios	1583703.92
4	moveis_decoracao	1429063.39
5	relogios_presentes	1427210.78

Analysis II Q5	Recommendations
FINDING	The top 5 categories are • cama_mesa_banho : Bed, Table & Bath (home linens, bedsheets, towels, tablecloths) • beleza_saude: Beauty & Health • informatica_acessorios: Computer/IT Accessories • moveis_decoracao: Furniture & Decoration • relogios_presentes: Watches & Gifts
IMPLICATION	These are not high margin product categories, so we can say: Amazon Brazil is making money because of volume purchases. The above-mentioned products are frequently purchased items.
ACTION	If beauty/health and home essentials drive the most revenue then Amazon India should ensure these categories never go out of stock. • Prioritize warehouse space and fast-delivery zones for these categories • Stockouts here would hurt revenue much more than stockouts in high margin products. • Amazon India should actively recruit more sellers in beauty/health, home decor, and gifting categories. • Amazon India could get more return on marketing investment by promoting high-frequency categories, daily deals, festival-specific gifting, beauty subscription boxes • Amazon can bundle products within these categories to increase basket size (e.g., "Buy bedsheet + pillow covers together", "Buy 3 skincare items, save 15%")

Analysis - III

#1 The marketing team wants to compare the total sales between different seasons.
Use a subquery to calculate total sales for each season (Spring, Summer, Autumn, Winter) based on order purchase dates, and display the results.

- **Spring is in the months of March, April and May.**
- **Summer is from June to August and**
- **Autumn is between September and November and**
- **rest months are Winter.**

Output: season, total_sales

Query | Query History

```
195
196 select
197     case
198         when extract(month from o.order_purchase_timestamp) between 2 and 6
199         then 'Spring'
200         when extract(month from o.order_purchase_timestamp) between 5 and 9
201         then 'Summer'
202         when extract(month from o.order_purchase_timestamp) between 8 and 12
203         then 'Autumn'
204         else 'Winter'
205     end as season,
206     sum(payment_value) as total_sales
207 from payments p
208 join orders o
209 on p.order_id = o.order_id
210 group by 1
211
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 4

	season text	total_sales numeric
1	Spring	7754518.43
2	Summer	4088199.54
3	Winter	1253492.22
4	Autumn	2912661.93

Analysis III Q1	Recommendations
FINDING	Spring recorded the highest sales followed by Summer, Autumn and Winter.
IMPLICATION	One of the implications can be that spring marks as new beginnings (like new school classes), resulting in purchases of clothes, stationery, books. Similarly winter marks the holiday season, some of the areas are snow covered in winter and there are major logistics challenges to fulfil the orders, resulting in longer delivery timelines and less order fulfilment.
ACTION	Amazon should plan its inventory seasonally, should stock up on school accessories before Spring and run back-to-school promotions. To increase sales in winters, Amazon should pre-stock inventory in local warehouses, adjust delivery timings, and offer early Winter deals.

#2 The inventory team is interested in identifying products that have sales volumes above the overall average. Write a query that uses a subquery to filter products with a total quantity sold above the average quantity.
Output: product_id, total_quantity_sold

Query | Query History

```
215
216  with cte as
217  ( select product_id, count(order_id) as total_quantity_sold
218  from order_items
219  group by product_id
220  )
221
222  select product_id, total_quantity_sold
223  from cte
224  where total_quantity_sold> (select round(avg(total_quantity_sold)) from cte)
225  order by total_quantity_sold desc
226
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 1000

	product_id character varying	total_quantity_sold bigint
1	aca2eb7d00ea1a7b8ebd4e6831466...	527
2	99a4788cb24856965c36a24e339b6...	488
3	422879e10f46682990de24d770e7f...	484
4	389d119b48cf3043d311335e499d9...	392
5	368c6c730842d78016ad823897a37...	388
6	53759a2ecddad2bb87a079a1f1519f...	373
7	d1c427060a0f73f6b889a5c7c61f2a...	343
8	53b36df67ebb7c41585e8d54d6772...	323
9	154e7e31ebfa092203795c972e580...	281
10	3dd2a17168ec895c781a9191c1e95...	274
11	2b4609f8948be18874494203496bc...	260
12	7c1bd920dbdf22470b68bde975dd3...	231
13	a62e25e09e05e6faf31d90c6ec1aa3...	226
14	5a848e4ab52fd5445cdc07aab1c40...	197

Total rows: 6366 | Query complete 00:00:00.154

#3 To understand seasonal sales patterns, the finance team is analysing the monthly revenue trends over the past year (year 2018).
Run a query to calculate total revenue generated each month and identify periods of peak and low sales.
Export the data to Excel and create a graph to visually represent revenue changes across the months.
Output: month, total_revenue

Query | Query History

```
-- #3 To understand seasonal sales patterns, the finance team is analysing the monthly revenue trends over the past year (year 2018).
-- Run a query to calculate total revenue generated each month and identify periods of peak and low sales.
-- Export the data to Excel and create a graph to visually represent revenue changes across the months.

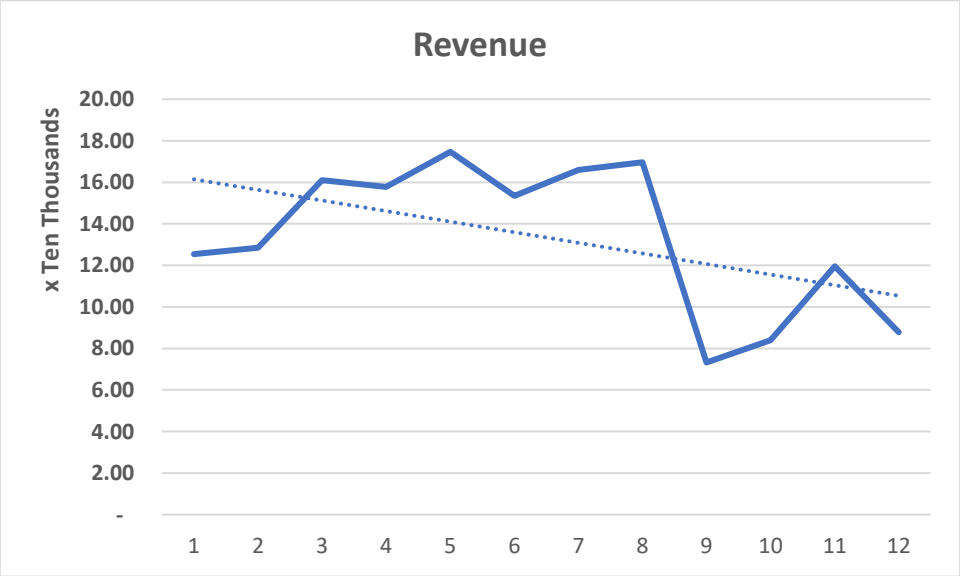
-- Output: month, total_revenue
select
extract (month from o.order_purchase_timestamp) as month, sum(p.payment_value) as revenue
from orders o
join payments p
on o.order_id = p.order_id
group by 1
order by 1 asc
```

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	month numeric	revenue numeric
1	1	1253492.22
2	2	1284371.35
3	3	1609515.72
4	4	1578573.51
5	5	1746900.97
6	6	1535156.88
7	7	1658923.67
8	8	1696821.64
9	9	732454.23
10	10	839358.03
11	11	1194882.80
12	12	878421.10

Total rows: 12 | Query complete 00:00:00.358 | CRLF



#4 A loyalty program is being designed for Amazon India.
Create a segmentation based on purchase frequency: 'Occasional' for customers with 1-2 orders, 'Regular' for 3-5 orders, and 'Loyal' for more than 5 orders.
Use a CTE to classify customers and their count and generate a chart in Excel to show the proportion of each segment.
Output: customer_type, count

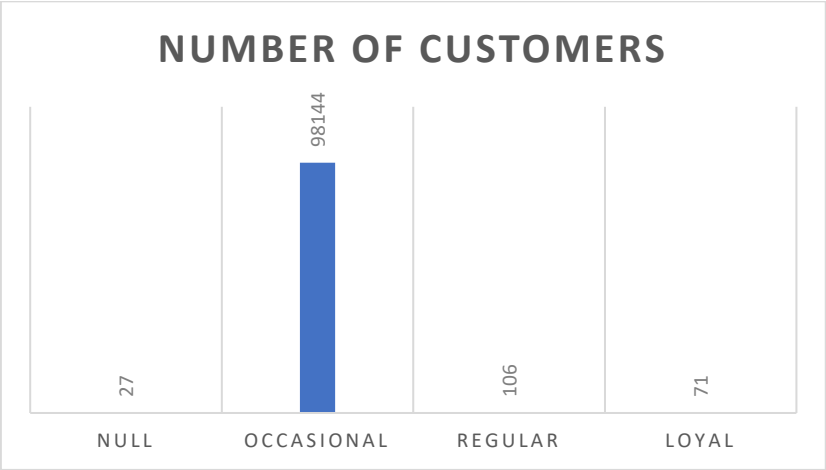
Query | Query History

```
250
251  with cte as
252  (
253      select
254          customer_id,
255          count(order_id) as purchase_frequency,
256          case
257               when count(order_id)<=2
258               then 'Occasional'
259               when count(order_id)<=5
260               then 'Regular'
261               when count(order_id)>6
262               then 'Loyal'
263          end as segment
264          from orders
265      group by customer_id
266      order by 1 desc
267  )
268
269  select segment, count(customer_id) as number_of_customers
270      from cte
271  group by segment
272
```

Data Output | Messages | Notifications

Showing rows: 1 to 4

	segment text	number_of_customers bigint
1	[null]	27
2	Occasion...	98144
3	Regular	106
4	Loyal	71



#5 Amazon wants to identify high-value customers to target for an exclusive rewards program.
You are required to rank customers based on their average order value (avg_order_value) to find the top 20 customers.
Output: **customer_id, avg_order_value, and customer_rank**

Query | Query History

```
273
274  -- #5 Amazon wants to identify high-value customers to target for an exclusive rewards program.
275  -- You are required to rank customers based on their average order value (avg_order_value) to find the top 20 customers.
276
277  -- Output: **customer_id, avg_order_value, and customer_rank**
278
279  select
280      o.customer_id,
281      round(avg(p.payment_value),2) as average_order_value,
282      dense_rank() over(order by avg(p.payment_value) desc) as customer_rank
283  from orders o
284  join payments p
285  on o.order_id = p.order_id
286  group by o.customer_id
287  order by 2 desc
288  limit 20
289
```

Data Output | Messages | Notifications

Showing rows: 1 to 20 | Page No: 1

	customer_id character varying	average_order_value numeric	customer_rank bigint
1	1617b1357756262bfa56ab541c47...	13664.08	1
2	ec5b2ba62e574342386871631fafd...	7274.88	2
3	c6e2731c5b391845f6800c97401a4...	6929.31	3
4	f48d464a0baaea338cb25f816991a...	6922.21	4
5	3fd6777bbce08a352fddd04e4a7cc...	6726.66	5
6	05455dfa7cd02f13d132aa7a6a972...	6081.54	6
7	df55c14d1476a9a3467f131269c24...	4950.34	7
8	e0a2412720e9ea4f26c1ac985f6a7...	4809.44	8
9	24bbf5fd2f2e1b359ee7de94defc4a...	4764.34	9
10	3d979689f636322c62418b6346b1...	4681.78	10

Total rows: 20 | Query complete 00:00:00.762

#6 Amazon wants to analyze sales growth trends for its key products over their lifecycle.
Calculate monthly cumulative sales for each product from the date of its first sale.
Use a recursive CTE to compute the cumulative sales (total_sales) for each product month by month.
Output: product_id, sale_month, and total_sales

Query | Query History

```sql
-- #6 Amazon wants to analyze sales growth trends for its key products over their lifecycle.
-- Calculate monthly cumulative sales for each product from the date of its first sale.
-- Use a recursive CTE to compute the cumulative sales (total_sales) for each product month by month.

-- Output: product_id, sale_month, and total_sales

with recursive product_lifecycles as (
    select
        oi.product_id,
        DATE_TRUNC('month', min(o.order_purchase_timestamp)) as sale_month,
        DATE_TRUNC('month', max(o.order_purchase_timestamp)) as max_month
    from order_items oi
    join orders o on oi.order_id = o.order_id
    group by oi.product_id

        union all

    select
        product_id,
        (sale_month + INTERVAL '1 month')::timestamp as sale_month,
        max_month
    from product_lifecycles
    where sale_month < max_month
        ),
    monthly_actual_sales as (

    select
        oi.product_id,
        DATE_TRUNC('month', o.order_purchase_timestamp) as sale_month,
        sum(oi.price) as monthly_revenue
    from order_items oi
    join orders o on oi.order_id = o.order_id
    group by oi.product_id, DATE_TRUNC('month', o.order_purchase_timestamp)
    )
```

Total rows: 98272 | Query complete 00:00:02.324

Query | Query History

```
323     )
324
325 select
326     pl.product_id,
327     TO_CHAR(pl.sale_month, 'YYYY-MM') as sale_month,
328     sum(coalesce(mas.monthly_revenue, 0)) over (
329         partition by pl.product_id
330         order by pl.sale_month
331     ) as total_sales
332 from product_lifecycles pl
333 left join monthly_actual_sales mas
334     on pl.product_id = mas.product_id
335    and pl.sale_month = mas.sale_month
336 order by pl.product_id, pl.sale_month
337
```

Data Output | Messages | Notifications

SQL

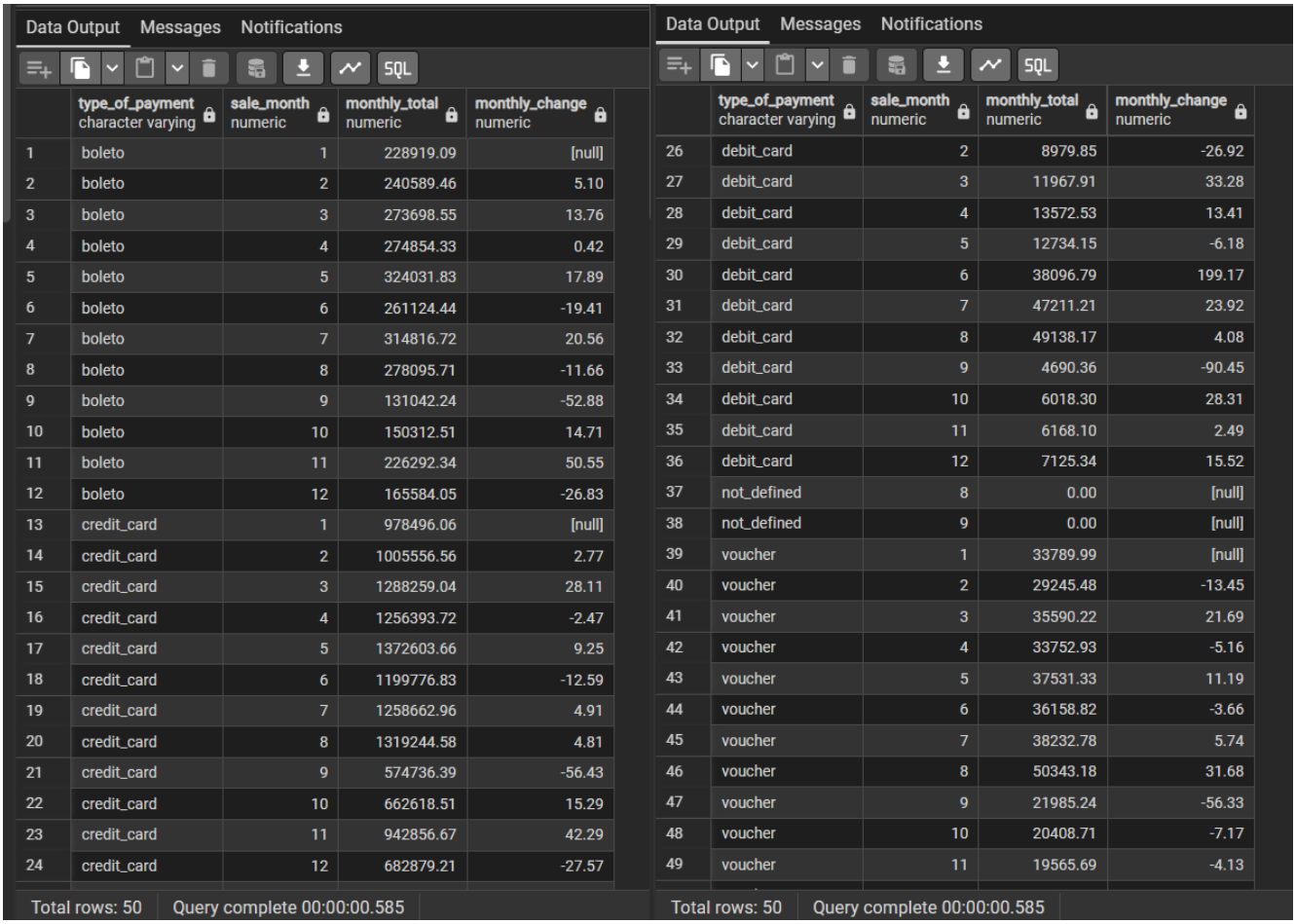
	product_id character varying	sale_month text	total_sales numeric
1	00066f42aeeb9f3007548bb9d3f33c38	2018-05	101.65
2	00088930e925c41fd95ebfe695fd2655	2017-12	129.90
3	0009406fd7479715e4bef61dd91f2462	2017-12	229.00
4	000b8f95fcb9e0096488278317764d...	2018-08	117.80
5	000d9be29b5207b54e86aa1b1ac548...	2018-04	199.00
6	0011c512eb256aa0dbbb544d8dffcf6e	2017-12	52.00
7	00126f27c813603687e6ce486d909d...	2017-09	498.00
8	001795ec6f1b187d37335e1c470476...	2017-10	38.90
9	001795ec6f1b187d37335e1c470476...	2017-11	116.70
10	001795ec6f1b187d37335e1c470476...	2017-12	350.10

Total rows: 98272 | Query complete 00:00:02.407

#7 To understand how different payment methods affect monthly sales growth; Amazon wants to compute the total sales for each payment method and calculate the month-over-month growth rate for the past year (year 2018).
Write query to first calculate total monthly sales for each payment method, then compute the percentage change from the previous month.
Output: payment_type, sale_month, monthly_total, monthly_change.

Query | Query History

```
338
339
340 -- #7 To understand how different payment methods affect monthly sales growth,
341 -- Amazon wants to compute the total sales for each payment method and
342 -- calculate the month-over-month growth rate for the past year (year 2018).
343 -- Write query to first calculate total monthly sales for each payment method, then compute the percentage change from the previous month.
344
345 -- Output: payment_type, sale_month, monthly_total, monthly_change.
346
347 With
348 cte as
349 (
350     select
351         p.payment_type as type_of_payment,
352         extract(month from o.order_purchase_timestamp) as sale_month,
353         sum(p.payment_value) as monthly_total
354     from payments p
355     join orders o
356     on p.order_id = o.order_id
357     group by p.payment_type, extract(month from o.order_purchase_timestamp)
358     order by 1
359 )
360 select
361     type_of_payment,
362     sale_month,
363     monthly_total,
364     round((monthly_total-lag(monthly_total) over(partition by type_of_payment order by sale_month))/
365     nullif(lag(monthly_total) over(partition by type_of_payment order by sale_month),0)*100,2) as monthly_change
366
367 from cte
368     order by 1,2
369
370
```



Data Output | Messages | Notifications

	type_of_payment character varying	sale_month numeric	monthly_total numeric	monthly_change numeric
1	boleto	1	228919.09	[null]
2	boleto	2	240589.46	5.10
3	boleto	3	273698.55	13.76
4	boleto	4	274854.33	0.42
5	boleto	5	324031.83	17.89
6	boleto	6	261124.44	-19.41
7	boleto	7	314816.72	20.56
8	boleto	8	278095.71	-11.66
9	boleto	9	131042.24	-52.88
10	boleto	10	150312.51	14.71
11	boleto	11	226292.34	50.55
12	boleto	12	165584.05	-26.83
13	credit_card	1	978496.06	[null]
14	credit_card	2	1005556.56	2.77
15	credit_card	3	1288259.04	28.11
16	credit_card	4	1256393.72	-2.47
17	credit_card	5	1372603.66	9.25
18	credit_card	6	1199776.83	-12.59
19	credit_card	7	1258662.96	4.91
20	credit_card	8	1319244.58	4.81
21	credit_card	9	574736.39	-56.43
22	credit_card	10	662618.51	15.29
23	credit_card	11	942856.67	42.29
24	credit_card	12	682879.21	-27.57

Total rows: 50 | Query complete 00:00:00.585

Data Output | Messages | Notifications

	type_of_payment character varying	sale_month numeric	monthly_total numeric	monthly_change numeric
26	debit_card	2	8979.85	-26.92
27	debit_card	3	11967.91	33.28
28	debit_card	4	13572.53	13.41
29	debit_card	5	12734.15	-6.18
30	debit_card	6	38096.79	199.17
31	debit_card	7	47211.21	23.92
32	debit_card	8	49138.17	4.08
33	debit_card	9	4690.36	-90.45
34	debit_card	10	6018.30	28.31
35	debit_card	11	6168.10	2.49
36	debit_card	12	7125.34	15.52
37	not_defined	8	0.00	[null]
38	not_defined	9	0.00	[null]
39	voucher	1	33789.99	[null]
40	voucher	2	29245.48	-13.45
41	voucher	3	35590.22	21.69
42	voucher	4	33752.93	-5.16
43	voucher	5	37531.33	11.19
44	voucher	6	36158.82	-3.66
45	voucher	7	38232.78	5.74
46	voucher	8	50343.18	31.68
47	voucher	9	21985.24	-56.33
48	voucher	10	20408.71	-7.17
49	voucher	11	19565.69	-4.13

Total rows: 50 | Query complete 00:00:00.585

Analysis III Q7	Recommendations
FINDING	• There is no certain pattern of increase or decrease in monthly sales for any payment type. • However, the sharpest increase in sales was around 199% during June and the sharpest decrease was around 90% during September, both attributed to debit card. • All the payment types had sharp dip in sales during the month of September. • All except debit card were showing negative growth in sales during the month of June. • Voucher had significant negative growth during 5 months of the year.
IMPLICATION	• It implies the usage of debit card for making payments in not consistent. The users of Dr card are occasional users who only use it when no other options are available. • The month of September registered comparatively less sales, which might be some cultural, economic or seasonal factor (yet to be explored) rather than a payment specific issue. • Vouchers are triggered externally, people use them when amazon issues them, so less use of vouchers in 5 months might mean comparatively less vouchers were issued by amazon during those months.
ACTION	• If Amazon wants to promote the use of debit card, so it can incentivise its use by giving cash backs or discounts. • It should plan inventory accordingly knowing there will be a sales dip in September. • Amazon can plan on issuing vouchers for promotions.